

6 (**long**) Figure 6 shows a composite beam comprising a steel I-beam and a concrete slab. The concrete is connected to the steel beam by a set of steel “shear studs” as illustrated. The dimensions given in the figure (which is not to scale) are given in cm. The steel beam has cross-sectional area 60 cm^2 and second moment of area $10,000 \text{ cm}^4$ about its own neutral axis. The Young’s modulus of steel and cement are 450 GPa and 30 GPa respectively.

(a) Calculate the bending stiffness, EI , of the composite beam. [20]

(b) If the beam is subjected to a vertical shear force of 100 kN and each shear stud can withstand a maximum shear force of 50 kN, what is the maximum permissible stud spacing, s , to prevent slippage between the two components of the beam? [10]

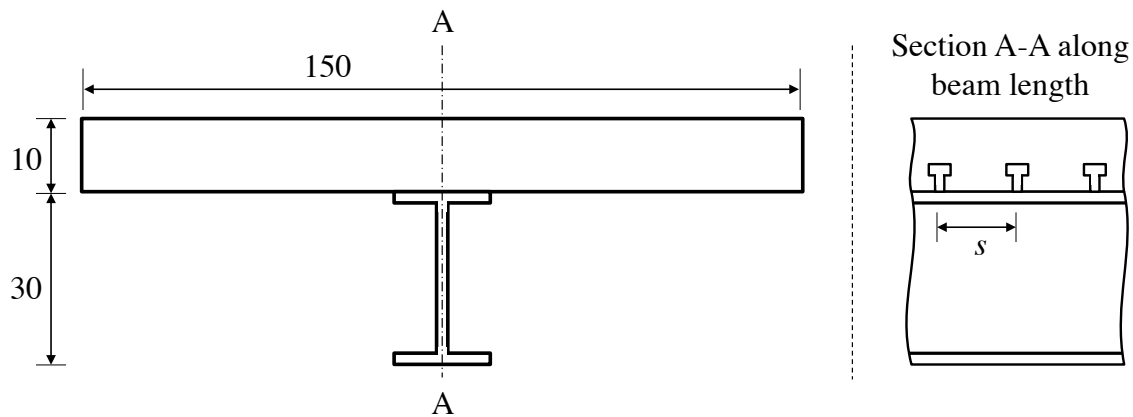


Fig. 6